Do LLMs forecast aftershocks like the USGS? A contamination-aware benchmark

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Introduction

After a large earthquake, people ask chatbots how many more aftershocks to expect. The US Geological Survey (USGS) answers the same question with operational aftershock forecasts [2, 3]. I asked whether general-purpose language models, given exactly what USGS knew at each forecast time, forecast as well, on earthquakes that began after the models' training cutoff, so the answers cannot have been memorized [5].

Results

I built forecast questions from the ComCat catalog and the archived official forecasts of eleven M≥7 sequences (July 2025 to October 2026): 61 forecast times, each asking for the number of M≥5 aftershocks in the next day, week and month. Each prompt held only the catalog as it stood at that time, rebuilt from ComCat's revision histories (it reproduces 95.1% of the official forecasts' own counts). Claude Haiku, Sonnet and Opus 5.5 answered 3,710 requests with nine quantiles or with Reasenberg-Jones parameters [1], with and without the catalog. I scored them with the log-scale quantile score [4] beside the official forecast, a refit of the USGS ETAS model and simpler baselines. Hypotheses and analysis were frozen and their SHA-256 posted publicly before any scored call (tag `prereg-v1`).

On the 33 settled windows of the two post-cutoff sequences with an official forecast (Colombia and Flores Sea), Opus and Sonnet could not be distinguished from USGS (score difference −0.022, 95% interval −0.074 to 0.032, and −0.015, −0.081 to 0.017; negative favours the model), while Haiku was worse (+0.124, 0.004 to 0.278; Figure 1b). With only two earthquakes the test cannot tell these top forecasters apart: the ETAS refit and even a naive rule of thumb (the aftershock count so far, extended with the usual decay) also tie USGS. The benchmark does separate weaker forecasters. Two open models run locally, gpt-oss-20b and Olmo 3.1 32B, were far behind, with 12% and 54% of answers unusable against at most 2.7% for Claude; fine-tuning gpt-oss-20b on ETAS forecasts for 515 earlier sequences (Adaption) made every answer usable but still trailed USGS (+0.14). Asking for model parameters instead of quantiles made every model worse, and showing the catalog helped. USGS and Opus both under-predicted the early Flores Sea aftershocks (Figure 1a).

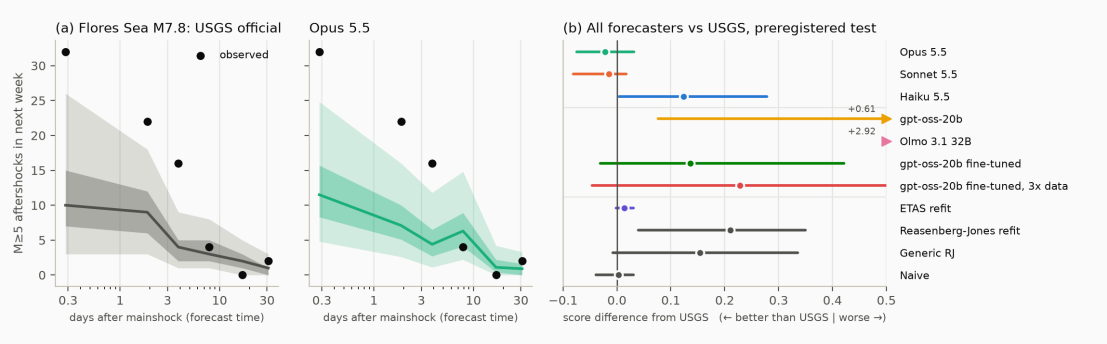


Figure 1. (a) Forecasts of the number of M≥5 aftershocks in the next week after the M7.8 Flores Sea earthquake, by USGS and Opus 5.5 (bands: 50% and 90% ranges; dots: observed). (b) Mean score difference from the USGS forecast on the 33 preregistered windows, with 95% moving-block bootstrap intervals; left of zero favours the forecaster.

After the freeze I scored 39 more sequences with an official forecast (M≥5, 18 after the Claude cutoff; exploratory, frozen code). Opus again matched USGS (+0.002, −0.002 to 0.007), and Sonnet without the catalog was clearly worse for M≥4 counts (+0.13). Claude did no better before its cutoff than after, and the open models, trained before both sets, kept the same gap on both: no sign of memorization. A fixed literature search (Valency) found no paper on any mainshock before its forecasts. Live forecasts for the ongoing Panama sequence were hashed and published before their windows opened; in the first closed window (provisional) Opus scored 0.015 against USGS's 0.027.

Future work

The benchmark grows with every new forecast-bearing sequence, which will give it the power to tell the top forecasters apart; next I will add tool-using models.

Open-source materials

Code, data, prompts, preregistration and every model output (tags `prereg-v1`, `scoring-v1`, `live-panama-1`, `live-panama-2`): https://github.com/tjnewton/llm-aftershock-forecast. Interactive explorer (marimo): https://tjnewton.github.io/llm-aftershock-forecast/. Video: https://youtu.be/qpPUB_VTlBY.

References

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